

Instructor: Date:

Learner: Regional Branch:

Program: Center:

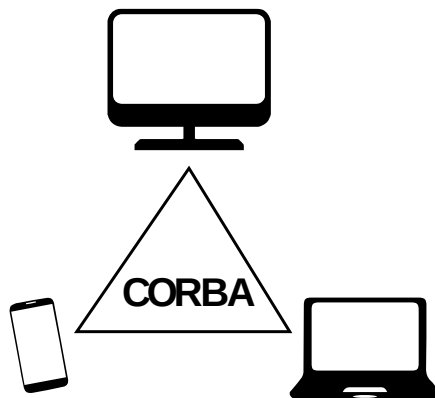
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Let's Start!

CORBA and Its Importance in Programming



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.

Explain important features related to object programming in different contexts and with different types of object programming.

Can Do Statements:

- I can explain what CORBA means.
- I can identify the importance of CORBA in software programming.
- I can describe briefly how CORBA works.

Ask your partner:

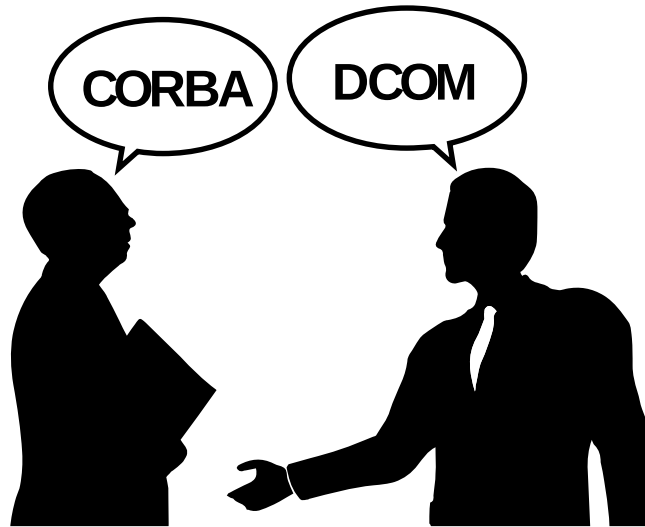
- Do you use different programming languages for your applications? Which ones?
- What is the worst experience you have had with programming languages? Share it with your classmates.





Let's Explore!

Listen to the professor talking about CORBA.



Instructions: Read the text below.

Professor: Today, we're going to talk about CORBA. If you know what CORBA is, raise your hand. Oh, ok. So, I'll explain from the beginning then.

CORBA is an acronym that stands for Common Object Request Broker Architecture, this is a standard used for creating and managing distributed program objects in a network. In other words, it allows programs to develop with different operating systems and with different locations to communicate through an "interface broker".

Student: Is that the main reason why it is so important?

Professor: Yes. There are many reasons, but the most important one is that CORBA improves the flexibility and portability of the applications. CORBA allows the programmer to work easily without having to go through difficult tasks caused by the heterogeneous computing environment. That's why there aren't many complaints about CORBA. If you use CORBA, you can work with different operating systems, programming languages, and computing hardware.

Student: And, how does CORBA work?

Professor: There is a process where the client program requests services from a server program or object, so the client doesn't need to understand where the server is or what the interface to the server program's looks like. When there are requests or the need to return a reply, programs use the General INTER-ORB protocol. Are there any questions?

Well, there isn't much time left. Tomorrow we will keep explaining everything you need to know about CORBA. If you don't understand something, bring your questions tomorrow so you can pass the test with flying colors.





Let's Practice!

Can you identify the most important ideas of CORBA?

Instructions: Read the text. Select all the correct answers.

There are many reasons why CORBA is important at the moment of programming an app. CORBA requests information from other servers, due to the fact that programmers don't know the language the other sever works with. It connects both, client and server, despite the language they were programmed. This is done by using an Internet Inter-ORB Protocol which eases the work of programmers. Even though, Microsoft works with the Distributed Component Object Model Architecture, Object Management Group, also called OMG, and Microsoft have an agreement in which both CORBA and DCOM can communicate each other without any restriction.

According to the audio, which of the following ideas describe the importance of CORBA?

1. ____ There are only some languages CORBA is compatible with.
2. ____ There are no language barriers, so CORBA connects clients and servers despite the programming language.
3. ____ There are two protocols CORBA uses to connect with remote servers.
4. ____ There is an agreement between Microsoft and OMG to connect the communication between their protocols.
5. ____ There is no need for programmers to worry about the remote server's programming language.



Let's Practice!

Let's check some basic concepts about CORBA.

Instructions: Match the sentences from Column A with Column B.

COLUMN A

1. If you use CORBA,
2. If you are programming an app, there are some protocols and rules
3. If you are talking about CORBA,

COLUMN B

- a. you are working with middleware applications.
- b. you can work with different operating systems.
- c. you need to know, such as CORBA that allows programs to communicate with remote servers with ease.



Extension Activity

Now, It's your turn to show how much you know about CORBA

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Do some further research about other middleware applications similar to CORBA
- Make a list where you write similarities and differences between them.
- Share those differences and similarities to your classmates and instructor.

Self-Assessment

Yes

No

- I do research on specific topics.
- I find similarities and differences between topics
- I share my findings to my classmates

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Target Words

- hardware** (noun): the mechanical devices that are part of a computer system
e.g. You use CORBA, you can work with different operating systems, programming languages, and computing hardware.
- look like** (phrasal verb): to resemble or be similar to something
e.g. The client doesn't need to understand where the server is or what the interface to the server program looks like.
- pass with flying colors** (idiomatic expression): to succeed or get a good grade in a test or exam
e.g. If you don't understand something, bring your questions tomorrow so you can pass the test with flying colors.
- request** (verb): to ask for something
e.g. When there are requests or the need to return a reply, programs use the General INTER-ORB protocol
- broker** (noun): an entity that functions as intermediary between two parties
e.g. It allows programs developed with different operating systems and with different locations to communicate through an "interface broker".
- reason** (noun): the motive for doing something
e.g. There are many reasons, but the most important one is that CORBA improves the flexibility and portability of the applications.



Learning Points

There Is, There Are in All Forms

Use **there is** and **there are** to talk about the things that you can see and the things you know that exist.

You use **there is** to talk about singular or uncountable nouns that you can see or touch. You use **there are** for plural countable nouns when you want to refer to its existences or presence.

Affirmative statements

There is some vital **information** in that flash drive. (uncountable noun)

There is a model we can use on diverse platforms. (singular noun)

There are many models we can use when programming. (plural noun)

Negative statements

There isn't or **there is no** vital information in that flash drive. (uncountable noun)

There isn't or **there is no** a model we can use on diverse platforms. (singular noun)

There aren't or **there are no** many models we can use when programming. (plural noun)

Question form

Is there vital information in that flash drive? (uncountable noun)

Yes, there is.

No, there isn't.

Is there a model we can use on diverse platforms? (singular noun)

Yes, there is.

No, there isn't.

Are there many models we can use when programming? (plural noun)

Yes, there are.

No, there aren't.

Examples

- *There are many reasons, but the most important one is that CORBA improves the flexibility and portability of the applications.*
- *That's why there aren't many complaints about CORBA.*
- *There is a process where the client program requests services from a server program or object.*
- *Well, there isn't much time left.*





Learning Points

Zero Conditional Review

Use the **Zero conditional to talk about cause and consequence**

You use the **zero conditional** to show the result of a possible action, this result always happens if you perform the action you just mentioned.

To form **affirmative sentences** with the zero conditional use an "if clause" and a "result clause" with two present simple verbs. Follow this structure:

If + subject + present simple, subject + present simple + complement

For example:

If you **use** OOP, the programming model **centers** in the objects and not the mathematical logic.

For **negative statements** there can be a negative result or a negative situation, so you can form it with two different possibilities:

The "if clause" in negative

For example:

If we **don't know** many programming languages, we **use** the ones we know to program.

The "result clause" in negative

For example:

If she **uses** only one model, she **can't work** with other platforms unless she uses CORBA.

Examples

- If you know what CORBA is, raise your hand.
- If you use CORBA, you can work with different operating systems, programming languages, and computing hardware.
- If you don't understand something, bring your questions tomorrow so you can pass the test with flying colors.





Learning Points

What is an acronym?

An **acronym** is a name or a word that you form by putting together the initials of a phrase or words.

You use **acronyms** as a way to abbreviate and standardize names, some acronyms are used to make it easier to talk about governmental entities or international organizations. Also, you can find some examples of acronyms in software programming,:

OMG: Object Management Group

ORB: Object Request Broker

TPC: Transmission Control Protocol

DCOM: Distributed Component Object Model

Examples

- *CORBA is an acronym that stands for Common Object Request Broker Architecture, this is a standard used for creating and managing distributed program objects in a network.*

